CMSC 199.2 (Luceres)

Comments from the Final Defense

1. Proper citation in presentation
2. Footnote for abbreviation in presentation
3. Fix panel name to "Samson Jr. D. Rollo"
4. 15 x 20 to 20 x20 shift explanation in presentation
5. Improve figure sizes in manuscript
6. Explain Well classification accuracy vs financial profitability and include sources
7. Improve justification for image-based encoding of time series data and add more sources
8. Implication of placement of MIC in the data. Are there effects? Provide sources on CNN data representation effects
9. Figures and manuscript are very small
10. Accuracy vs Profitability – show how model performed on period where stocks were highly volatile (i.e., during 2021 pandemic)
11. Must explicitly clarify in the manuscript that the 11-day future windows were strictly used for ground truth labels during the training phase.

CMSC 199.2 (Macadine)

Comments from the Final Defense

1. Correct panel name "Samson Jr. D. Rollo"
2. Improve YOLOv8 figure size
3. Add highlight in the results which numbers outperforms the rest in the table in manuscript
4. Fix pp. 49-50, 51-53 and the rest with figures
5. If time permits, test over mask R-CNN.
6. Include statistical validation gains and testing
7. Add failure analysis to give light on what the model is missing
8. Occluded pollens mentioned in problem statement does not address it directly for instance separation discussion in the results about the findings
9. Justify performance over complexity of combined CLAHE + EFF. What is the reason behind? Provide sources for justification.
10. Mathematical justification of the tradeoff of speed vs recall (e.g. CLAHE is faster, while EFF is longer but yields better recall
11. Combining CLAHE and EFF actually decreased accuracy. Should add a dedicated paragraph in the manuscript explaining why this happened.
12. Paper implies that image enhancement solves occlusion. Image enhancements fixes lighting, not physical overlap.

CMSC 199.2 (Dalomias)

Comments from the Final Defense

1. Fix panel name "Samson Jr. D. Rollo"
2. Add highlights to the table to indicate best result on that field
3. Add cross validation or external dataset tests
4. Add sources on quantification of melanin counter and/or annotation sensitivity impact
5. Include statistical variance in result
6. If time permits, add other baseline model comparison
7. Justify small dataset over results, best to use sources/studies.
8. Image dataset is small and deep learning models are highly prone to overfitting on dataset this small.
9. Implement *k*-fold cross validation to prove that the result was not just because of "lucky" train-test split